Photovoltaic cells use semiconductor material to turn incoming light into direct-current electricity. Sunlight frees electrons in the cell, creating an electric current. A group of connected cells forms a solar module or panel.

An inverter converts the direct current produced by solar panels into alternating current. Household appliances and the public electricity grid use alternating current. The inverter also monitors the system and may disconnect it during a grid outage.

Solar irradiance measures the power of sunlight received per unit area. The direction a panel faces and its tilt angle affect how much radiation it captures. Clouds, season, and time of day also change the available sunlight.

A microinverter is attached to an individual solar panel and converts that panel's direct current separately. This can reduce the effect of shade on neighboring modules. A conventional string inverter connects several panels to one shared conversion unit.

A home battery stores surplus electricity for later use, such as after sunset or during a brief outage. Its round-trip efficiency compares the energy recovered with the energy used to charge it. Storage capacity is measured in kilowatt-hours.

A wind turbine converts the kinetic energy of moving air into electricity. Its blades turn a rotor connected to a generator. Capacity factor compares the actual energy produced over time with the output it would produce at its rated power continuously.